Table 1. Pseudo Code for YIER-VLM Model Training

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Algorithm 1 YIER-VLM Training

Input : pre-trained vision-language model M0 (Qwen2.5-VL-7B-Instruct);
        domain document corpus D (medical laboratory reports, imaging reports,
        reimbursement documents, and general office documents)
Output: domain-adapted structured-extraction model M*

 1: for each document image x in D do
 2:     y ← verified key-value annotation of x in JSON format
 3:     s ← (x, natural-language extraction instruction, y)
 4:     D_sft ← D_sft ∪ {s}
 5: end for
 6: D_sft ← D_sft ∪ D_vqa          ▷ mix in a small amount of general VQA data to
                                    preserve general understanding ability
 7: freeze the vision encoder of M0
 8: attach lightweight low-rank adapters to the projector and the language decoder
 9: M_sft ← SupervisedFineTune(M0, D_sft)
                                   ▷ Stage 1: domain supervised fine-tuning; update
                                     only the adapter parameters
10: for each instruction sample s in D_sft do
11:     y_w ← verified annotation of s      ▷ preferred response
12:     y_l ← CorruptResponse(y_w)          ▷ rejected response: one of missing field,
                                              tampered value, or format error
13:     D_dpo ← D_dpo ∪ {(s.image, y_w, y_l)}
14: end for
15: M* ← PreferenceOptimize(M_sft, D_dpo)
                                   ▷ Stage 2: structured-output preference
optimization;
                                     improve key-field capture stability and output
consistency
16: return M*
```